

Shuqi Shen

+1 9842150748 | shuqishen27@gmail.com | linkedin.com/in/shuqi-shen-59a5302bb/

EDUCATION

Duke University

Master of Science, Electrical & Computer Engineering

Durham, NC

Aug 2023 - Dec 2025

Wuhan University of Technology

Bachelor of Engineering, Electrical Engineering

Wuhan, China

Sep 2019 - June 2023

AWARDS: National Scholarship, Outstanding Graduates

RELATED COURSES: Engineering Robust Server Software, System Programming, Operating System, Computer Network, Database System, Distributed System, Data Structure and Algorithms, Human-Centered Security and Privacy

SKILLS

Programming Language: Python, C++/C, GoLang, Javascript, Java, Swift, Verilog, MATLAB, SQL, HTML

Tools/Frameworks: Git, MVC, PostgreSQL, MySQL, RPC, Redis, Docker, Spring Framework, Django, Flask, React, Vue, AWS, TypeScript, Recommendation Infra, Emacs, Valgrind, GDB, Spark, Firebase, Xcode, SwiftUI, Agile, Linux Environment, CI/CD

Experience

Amazon, Software Engineer Intern

June 2025 – Aug 2025

- Developed an internal **full-stack web** application with a **Java-based backend service** and a **Java Spring**-powered web layer, using **TypeScript (React)** on the frontend to support real-time global logistics operations.
- Built an automated data sync service by integrating **AWS Lambda** and a backend **SQS consumer** to batch update 69K+ shipper sales contacts, ensuring consistency across **DynamoDB** and **Elasticsearch**.
- Built a full-stack feedback system: developed a configurable **React** component for collecting user ratings, and implemented a serverless pipeline (API Gateway → **Lambda** → **DynamoDB** → **EventBridge**-triggered export to **S3** → **QuickSight**) for daily aggregation and analysis of user experience data.

Volvo Cars, Software Engineer Intern

Sep 2024 – Dec 2024

- Led the development of **CI Hub**, a full-stack web application for internal employees to reserve, view, and manage rig information efficiently. Built a scalable backend with **Python** and **Flask**, delivering secure **RESTful APIs** for high-volume data transactions. Developed a responsive frontend using **Vue.js**, providing an intuitive user experience.
- Integrated **Azure Active Directory (AAD)** with OAuth to implement secure authentication, streamlining access through a reliable login workflow.

Kunlun Tech Co., Ltd., Software Engineer Intern

June 2024 – Sep 2024

- Developed the **recommendation system architecture**, improving efficiency by optimizing recall, rank, and rerank processes. Integrated the backend with ML model APIs to deliver input features, retrieve predictions, and execute **A/B testing**.
- Built a new content exposure dispersion strategy by implementing a **sliding window algorithm**, balancing **user experience** and increasing new content **click-through rate** by **11.92%** and **retention rate** by **9.77%**.
- Created a **system degradation module** that dynamically adjusted system complexity under **high loads**, enhancing **system stability** and reducing failure rates by approximately **20%**.
- Developed high-concurrency **GoLang microservices** with **gRPC** for inter-service communication, leveraging **goroutines** to improve throughput. Designed a tiered caching strategy with local cache, **Redis**, and **MySQL**, cutting data retrieval times and reducing database load.

Duke University, Software Engineer Intern (Remote)

May 2024 – Apr 2025

- Developed a phantom limb rehabilitation app and an oculomotor training app in **Swift** and **SwiftUI**, supporting video exercises, eye-tracking calibration, and progress tracking for patients in a North Carolina hospital.
- Implemented **MVVM** architecture and unit testing in **Xcode**, improving maintainability and reliability.
- Integrated **Firebase** for real-time data sync and designed a user-friendly **UI**, boosting engagement and adoption.

Projects

Http Cached Proxy (C++, Network, TCP Socket, Multi-Thread)

Jan.2024– Mar.2024

- Established a HTTP caching proxy server **daemon** which can handle GET, POST, CONNECT requests.
- Used **C++ 11** to implement **RAII** technique and modeled class with strong exception safety guarantee.
- Added **concurrency** to handle requests from different endpoints. Used **TCP sockets** to send and receive packets.
- Added RFC 7234-compliant conditional caching (ETag, Last-Modified) and LRU eviction, achieving **60%+ cache hit rate**.

Exchange Matching Engine (JAVA, Thread-Safe)

Nov.2023 - Dec.2023

- Developed a scalable matching engine in **Java** for stock and commodity markets to efficiently match buy and sell orders.
- Employed **pessimistic locking** (using **ReentrantLock**) and concurrency control to support partial fills, multiple fills, and order cancellations under high load, achieving a **40% improvement in system throughput** and ensuring correctness in multi-threaded environments.